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Apparatus and method for producing scored dough pieces

Abstract

Frozen un-proofed dough pieces are produced by cutting a dough sheet into strips and producing a score through an upper skin of each strip with at least one blunt deformation tool prior to freezing. The un-proofed dough pieces are formed by mass production, with an un-proofed dough sheet being cut to form dough strips which are then cut into the un-proofed dough pieces, with the score for each un-proofed dough piece being formed by pressing the upper skin of the dough down into the body of the dough, thereby deforming and pre-stressing the dough to create weak zones, either before, during or after the individual un-proofed dough pieces are actually created, but before freezing. Upon heating, openings are established in the upper skin along score lines.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2887964	12/1958	Griner	N/A	N/A
3225717	12/1964	Page	N/A	N/A
3765909	12/1972	Moline	N/A	N/A
4246838	12/1980	Pulver et al.	N/A	N/A
4276800	12/1980	Koppa et al.	N/A	N/A
4469476	12/1983	Cavanagh et al.	N/A	N/A
4650687	12/1986	Willard et al.	N/A	N/A
5349759	12/1993	Anton et al.	N/A	N/A
5935629	12/1998	Martin	N/A	N/A
6783782	12/2003	Larsen et al.	N/A	N/A
6902754	12/2004	Evans et al.	N/A	N/A
8535039	12/2012	Evans et al.	N/A	N/A
10238119	12/2018	Evans et al.	N/A	N/A
2003/0024360	12/2002	Ribble	N/A	N/A
2004/0197454	12/2003	Henry et al.	N/A	N/A
2005/0031733	12/2004	Domingues et al.	N/A	N/A
2005/0129821	12/2004	Goedeken	426/549	A21D 6/001
2007/0178208	12/2006	Moidl et al.	N/A	N/A
2008/0289466	12/2007	Takama et al.	N/A	N/A
2010/0159095	12/2009	Suski et al.	N/A	N/A
2010/0173052	12/2009	Suski et al.	N/A	N/A
2011/0177190	12/2010	Evans	425/294	A21C 11/10
2018/0289021	12/2017	Zorn	N/A	N/A
2019/0307135	12/2018	Gottenbusch et al.	N/A	N/A
2019/0357549	12/2018	Crosby	N/A	A21C 11/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3103810	12/2019	CA	N/A
19549762	12/2006	DE	N/A
1114890	12/1955	FR	N/A
2666490	12/1991	FR	N/A
WO 2004/037003	12/2003	WO	N/A
WO 2004/039163	12/2003	WO	N/A
WO 2015/160238	12/2014	WO	N/A

OTHER PUBLICATIONS

Ken Hagedorn, "Equipment Overview: A Complete Production Line for Artisan Bread"; Oct. 31, 2018 | Artisan Bread, Bakery Equipment obtained from <https://www.naegele-inc.com/2018/10/31/equipment-overview-a-complete-production-line-for-artisan-bread/> (Year: 2018). cited by examiner

J4 Tunnel Ovens, "*Baguette Line, Tray System*", Retrieved from the Internet: URL:<https://www.youtube.com/watch?v=tmr12pEBflag>, Feb. 26, 2021. cited by applicant

Ruth, "*Scoring Bread*", Cook Til Delicious, Retrieved from the Internet: <https://cooktildelicious.com/scoring-bread>, May 16, 2017. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention pertains to the art of food production and, more particularly, to the automated production of bread products. Bakery operators sometimes purchase frozen un-proofed bread dough, which they thaw, proof and bake prior to sale. Depending on the desired bread product, the bread dough may be manually scored after proofing and before baking to help control expansion during baking and to give the resulting bread product its traditional look. That is, by scoring after proofing, the dough is allowed to fully expand during the proofing process in an intact form and the scores are then provided in a decorative pattern just before baking to enable oven expansion in controlled regions. However, manually scoring each dough piece is time consuming and can actually represent an operational challenge. With this in mind, it would be beneficial to bakery operators if the frozen un-proofed dough already included a score. However, the changes that dough undergoes during proofing makes it difficult to utilize pre-proof scoring in a consistent manner, particularly for a commercial dough producing operation.

SUMMARY OF THE INVENTION

(2) The present invention successfully achieves the goal of scoring bread dough before purchase by bakery operators by employing deformation scoring. That is, a blunt deformation tool is employed to deform the dough, either before, after or simultaneously with a dough cutting operation, to pre-stress the dough in scoring regions to establish weak zones which open up along score lines upon cooking the dough. In accordance with preferred embodiments of the invention, after cutting and deformation scoring, the dough is frozen. More particularly, un-proofed dough pieces with scores are produced by directing a dough sheet along a conveyor with a lower skin of the dough sheet contacting the conveyor and an upper skin being spaced from the lower skin. At least one cutter cuts the dough sheet into separate dough strips. In one form of the invention, a series of rotating, circular blunt-edge dough cutters creates a series of strips entirely by pressing the upper skin toward the lower skin to pinch the upper and lower skins together while cutting. This technique stretches the upper skin of the dough as it is drawn toward the lower skin and pinches the two surfaces together as the dough is cut to form dough strips or pieces. Regardless of the type of dough cutters used to establish the strips, a blunt deformation tool is employed in accordance with the invention for the purpose of establishing one or more score lines in a respective strip upon cooking of the dough. Importantly, no actual cutting or slicing of the dough is performed for the scoring in accordance with the invention. Instead, a blunt deformation tool is used to pre-condition the dough in creating a functional score by pressing against and drawing the upper skin to a depth

in a body of the dough above the lower skin. After the deformation tool is no longer pressing on the dough, the dough actually rebounds or re-expands in the scoring region(s) while weak zones are established. The strips are cut into pieces which can be frozen and then later baked to make loaves of bread. It has been found that such a created score pre-stresses the dough in such a way that the dough stays intact while the dough piece goes through the proofing process but still causes a desired split to open when the piece is baked.

(3) In certain preferred forms of the invention, both the cutting of strips and the scoring are performed by a plurality of alternating blades and wheels fixed to a common driveshaft. In another embodiment, the scoring is performed by a plurality of wheels fixed to a first driveshaft, and the cutting (blunt-edge or otherwise) is performed by a plurality of blades fixed to a second driveshaft, with the first driveshaft located either upstream or downstream, i.e., longitudinally offset along the conveyor, of the second driveshaft. In accordance with this embodiment, the first driveshaft can be adjusted in the vertical direction to enable the plurality of wheels to create discontinuous scoring depressions or varying depth depressions along a longitudinal line by selectively raising or lowering the first driveshaft.

(4) Additional objects, features and advantages of the invention will become more readily apparent from the following detailed description of preferred embodiments thereof when taken in conjunction with the drawings wherein like reference numerals refer to common parts in the several views.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a portion of a production line for producing dough pieces in accordance with the present invention.

(2) FIG. 2A is a front view of a conveyor belt and combination rotary cutter and scorer unit of the production line without a dough sheet in place.

(3) FIG. 2B is a front view of the conveyor belt and combination rotary cutter and scorer unit of the production line with a dough sheet being cut and scored.

(4) FIG. 3 is a perspective view of the portion of the production line, showing a dough sheet being cut and scored by the combination rotary cutter and scorer unit.

(5) FIG. 4 is a perspective view of a dough piece formed using the production line.

(6) FIG. 5 is a perspective view of the dough piece after baking.

(7) FIG. 6 is a perspective view of a portion of a second production line for producing dough pieces in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(8) Detailed embodiments of the present invention are disclosed herein. However, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale, and some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to employ the present invention. Additionally, as used in connection with the present invention, terms such as “parallel”, “perpendicular”, “vertical”, “upper”, “lower” and the like do not necessarily require, for example, that the relevant items be perfectly parallel. Instead, these terms constitute general, relative directional terms which can vary within a reasonable margin of error or perspective so long as the error does not prevent the present invention from functioning as intended.

(9) As discussed above, it would be desirable to provide bread dough that is scored before delivery to bakery operators, preferably in a frozen un-proofed state. However, attempts have shown that it

can prove difficult to automate scoring of un-proofed dough pieces prior to freezing in a manner that consistently produces the desired end result upon baking. The present invention successfully achieves this goal in a simplified manner basically by performing the scoring of bread dough with a dough deformation technique that stretches the upper skin of dough to establish an elongated depression which, although significantly rebounding due to elasticity of the dough, pre-stresses the dough so as to establish a weak zone along the score line, with the dough splitting at the weak zone during baking to establish a desirable color differentiation between an outer crust and the exposed inner body in the region of the score.

(10) Various dough cutting approaches are described in U.S. Pat. No. 6,902,754, for example, which is incorporated herein by reference. This patent discloses a cutter including a blunt-edge dough-cutting or dough-engaging surface that is configured to draw a first or upper surface or skin of a dough toward a second or lower surface or skin of the dough when the cutter is impinged on the dough. The various dough cutting techniques described in U.S. Pat. No. 6,902,754 can be more generally referred to herein as “blunt-edge” cutting. Basically, this technique does not actually cut the dough in a conventional slicing operation as the cutting tool is blunt, but rather stretches the upper skin of the dough as it is drawn toward the lower skin and pinches the two surfaces together in cutting the dough to form dough strips or pieces. This advantageously results in a dough product that has rounded dimensions and more closely resembles a hand-formed dough product than dough cut by traditional techniques. Although not a required cutting technique in accordance with the overall invention, blunt-edge cutting of the dough into strips or pieces is preferred for the reasons mentioned above and will be described hereinafter in reference to exemplary embodiments of the invention.

(11) It has been discovered that establishing a score by pushing down the upper skin of the un-proofed dough prior to freezing results in an internal weak zone, resulting in a scored dough product with functional attributes comparable to dough products that are scored after proofing. Although there are multiple ways to cut a dough sheet into strips or pieces in accordance with the overall invention, scoring of the strips or pieces specifically employs blunt deformation technology, with the scoring being performed, from a manufacturing standpoint, just before, immediately after or at substantially the same time as cutting the dough into pieces.

(12) With reference to FIG. 1, an embodiment of the present invention is shown where blunt-edge cutting is employed, with the blunt-edge cutting and deformation scoring taking place simultaneously. Specifically, FIG. 1 shows a portion of a production line including a conveyor system **100** and a combination rotary dough cutter and scoring unit **105**. Conveyor system **100** includes a conveyor belt **110** for supporting and transporting a dough sheet (not shown in FIG. 1). However, other conveyor arrangements known in the art can be used with the present invention. As shown, conveyor belt **110** passes beneath rotary dough cutter and scoring unit **105** while traveling in a direction **115**.

(13) Rotary dough cutter and scoring unit **105** includes a driveshaft **120**, a plurality of circular blades **125-129** and a plurality of wheels **130-133**. Driveshaft **120** is aligned perpendicular to direction **115**. Accordingly, the axis of rotation **135** for rotary dough cutter and scoring unit **105** is also aligned perpendicular to direction **115**. Rotary dough cutter and scoring unit **105** is configured such that contact between blades **125-129** and conveyor belt **110**, or a dough sheet supported by conveyor belt **110**, causes rotary cutter **105** to rotate in a direction **140** as conveyor belt **110** travels in direction **115**. Alternatively, a motor and transmission (not shown) can be provided for driving rotary dough cutter and scoring unit **105**, i.e., for causing rotary dough cutter and scoring unit **105** to be positively driven to rotate in direction **140**.

(14) Blades **125-129** and wheels **130-133** are fixedly coupled to driveshaft **120** such that rotation of driveshaft **120** results in rotation of blades **125-133**. Each of blades **125-129** is configured to cut and each of wheels **130-133** is configured to score a dough sheet being transported by conveyor belt **110**. Specifically, blades **125-129** are blunt-edge blades or other circular cutters and are

configured to cut the dough sheet into dough strips, while wheels **130-133** are specifically configured to not cut or slice the dough sheet but rather uniquely score the dough sheet. For purposes of the present invention, “cutting” (and its variants) means that the dough is cut all the way through. This can be done in various ways, including cutting by pinching, as described above in connection with blunt-edge cutters, or cutting by slicing as done with sharp edge cutters. “Scoring” (and its variants) means that the dough is deformed only part of the way through. Specifically for purposes of this invention, “scoring” does not encompass “cutting” or “slicing” of the dough but rather “deforming” or “creasing” the dough is performed as will become more fully evident below. Also, while a select number of blades and wheels are shown across driveshaft **120**, other numbers of blades and interposed wheels can be used depending on the width of the dough sheet and the desired width of the resulting dough strips.

(15) Rotary dough cutter and scoring unit **105** is supported above conveyor belt **110** by a support structure **145**. The details of support structure **145** are not important to the present invention. However, it can be seen that support structure **145** includes identical left and right supports **150**. Each support **150** has a base **155** and a pair of columns **160**, supporting a housing **165**, which can incorporate a drive motor. Housings **165** receive the ends of driveshaft **120** to facilitate rotation of rotary cutter **105**.

(16) Turning to FIGS. 2A and 2B, a front view of conveyor belt **110** and rotary cutting and scoring unit **105** is provided. That is, the views of FIGS. 2A and 2B are taken in the direction opposite direction **115**. This view highlights the positioning of blades **125-129** and wheels **130-133** relative to conveyor belt **110**. Specifically, blades **125-129** are positioned in contact with, or just out of contact with, conveyor belt **110** so as to cut a dough sheet traveling along conveyor belt **110** (not shown in FIG. 2A). Again, in the preferred embodiment shown, each of blades **125-129** has a blunt-edge dough engaging portion (not separately labeled) designed to stretch the upper skin of the dough sheet and cut the dough as detailed further below. On the other hand, wheels **130-133** are positioned or spaced significantly further from conveyor belt **110** so as to only press the upper skin down into a body of the dough, thereby deforming the dough sheet. To accomplish this, blades **125-129** have greater diameters than wheels **130-133**. For example, wheels **130-133** have a diameter which is at least one-quarter ($\frac{1}{4}$) of an inch less than blades **125-129** such that wheels **130-133** do not reach conveyor belt **110**. In addition, as clearly shown, wheels **130-133** are significantly thinner than blades **125-129**, generally at least half as thick, more preferably in the order of about one-third ($\frac{1}{3}$) to one-fourth ($\frac{1}{4}$) the thickness.

(17) With reference now to FIG. 3, a dough sheet **300** has an exposed upper dough skin, a lower dough skin and a body therebetween (not separately labeled in this figure but discussed further below) supported on and transported by conveyor belt **110**. Although not shown, it should be understood that the dough of dough sheet **300** is preferably formed in a batch maker or the like and then processed into dough sheet **300**. Dough sheet **300** passes beneath rotary dough cutter and scoring unit **105**, which is configured to continuously cut and score dough sheet **300** as dough sheet **300** is transported in direction **115**. For example, FIGS. 2B and 3 show a plurality of dough strips **305**, which were formed from dough sheet **300** by blades **125-129**, and a plurality of scores **310**, which were formed in dough strips **305** by wheels **130-133**. In this exemplary embodiment, each score **310** extends parallel to a longitudinal axis of an associated dough strip **305**. Later, dough strips **305** are cut crosswise (i.e., in a direction substantially parallel to driveshaft **120**) to form a plurality of dough pieces.

(18) In particular, the dough engaging portion (not separately labeled but clearly shown in FIG. 2B) of each blade **125-129**, while also shaping the dough, draws the upper dough skin all the way to the lower dough skin to pinch the two skins together against the conveyor belt **110** while cutting. At the same time, first and second longitudinally extending dough strip sides (not separately labeled here but discussed further below) are formed (advantageously with rounded portions) from the upper dough skin for each dough strip or piece. Again, employing blunt-edge cutting in producing dough

strips **305** is preferred but not required in accordance with the present invention. Instead, this technique is encompassed by U.S. patent application Ser. No. 16/779,886 filed Feb. 3, 2020, and entitled “Apparatus and Method for Producing Scored Dough Pieces” which is incorporated herein by reference. However, by cutting the dough in this manner, the upper dough skin is advantageously stretched at or about the same time that wheels **130-133** score the dough, as detailed further below.

(19) Again, important in connection with the present invention is the scores are not created by cutting or slicing of the dough. Instead, the invention can be characterized to employ blunt deformation tools for wheels **130-133** to deform the dough, either before, after or simultaneously with a dough cutting operation, to pre-stress the dough in scoring regions to establish internal weak zones in the body of the dough which open up along score lines upon cooking the dough. Here, it should be noted that the term “wheels” is employed as being distinguishing from the term “blades” used herein since wheels **130-133** are not sharp and they do not perform any cutting or slicing of the dough in connection with creating the scores. Instead, wheels **130-133**, with their smaller diameters, draw or press the upper skin of the dough down to above the level of conveyor belt **110** (particularly see FIG. 2B) such that the upper skins of dough strips **305** never reach the lower skins (e.g., a quarter inch separation) due to the construction and operation of wheels **130-133**. With this operation, the upper skin is pressed into the body of the dough strips **305** so as to deform the body and stress the gluten matrix of the dough to initially establish elongated deep depressions in the dough. At this point, due to the elasticity of the dough, the dough rebounds to a significant extent, while internal weak zones remain in the dough, in the form of score **310**. In this sense, wheels **130-133** basically function to press the upper skins into inner body dough portions of strips **305**. Interestingly, it was found that the fairly deeply drawn upper skins do rebound a significant amount but a slight elongated depression (score), established by the deformation (physical change in the internal dough structure), remains in the dough before freezing and, as will be discussed further below, opens up upon cooking of the dough.

(20) One such cut dough piece which has at least partially rebounded following the scoring by deformation process of the invention is shown in FIG. 4. In particular, a dough piece **400** has an upper surface or skin **405**, a lower surface or skin **410** opposite upper surface **405**, opposing side surfaces **415** formed from the upper skin **405** and advantageously rounded due to the particular (blunt-edge) cutting arrangement, and opposing end surfaces **420**. Score line **310** was formed in upper surface **405** by one of wheels **130-133**, and side surfaces **415** were formed by two of blades **125-129**. End surfaces **420** were formed by another dough cutter (not shown) when one of strips **305** was cut crosswise. This other cutter can take the form of a guillotine dough cutter, for example. At this point, it should be pointed out that employing the scoring by deformation in combination with the blunt-edge cutting has a synergistic effect. That is, the blunt-edge cutting will inherently stretch upper skin **405** such that skin **405** is even more stretched and pre-stressed upon pressing skin **405** by a respective wheel **130-133** to create the weak zones during scoring. After formation, dough piece **400** is then frozen, preferably in an un-proofed state. At this point, it should be noted that “un-proofed” in accordance with the invention means that no substantial or intentional actions are taken to promote proofing. Rather, the dough pieces are, at the very least, frozen shortly after being formed so as to save proofing for later. Once frozen, un-proofed dough piece **400** can be transported to a bakery operator, for example, who thaws, proofs, bakes and sells the resulting bread product.

(21) FIG. 5 shows such a bread product. Specifically, FIG. 5 shows a bread product **500**, which corresponds to dough piece **400** after baking. As such, bread product **500** also has upper surface **405**, lower surface **410**, side surfaces **415** and end surfaces **420**. In addition, score **310** can be seen near end surfaces **420**. However, in the central portion of bread product **500**, score **310** has opened up through bulk expansion in the weak zone, forming an opening **502**. This automatically occurs during baking of dough piece **400**, due to the pre-stressing of dough piece **400** during scoring, as

dough piece **400** expands. In the embodiment illustrated, dough product **500** is in the form of a baguette. However, it should be recognized that other bread products can certainly be produced in accordance with the present invention. In any case, results have shown that, in accordance with the invention, a very consistent opening of the product along score **310** occurs upon baking. Therefore, the scoring by deformation is considered to be extremely advantageous to provide uniformity of the resulting products offered upon baking. In fact, the use of blunt deformation tools for scoring in accordance with the invention has been found to produce an improved visual appearance in the score area as the score opens up to a greater extent than prior cutting/slicing offerings during the baking process. This enhanced opening up of the product, e.g., baked loaf of bread, results in a greater visual contrast between the brown crust and the lighter interior of the overall product.

(22) With reference now to FIG. **6**, an embodiment of the present invention is shown where scoring takes place before blunt-edge cutting. Specifically, FIG. **6** shows a portion of a production line including conveyor system **100**, a rotary dough scorer **600** and a rotary dough cutter **605**. Conveyor system **100** is constructed and functions in the same manner as in the embodiment of FIG. **1**. That is, conveyor belt **110** of conveyor system **100** transports dough sheet **300** in direction **115** such that dough sheet **300** passes beneath rotary scorer **600** and rotary cutter **605**.

(23) Rotary scorer **600** and rotary cutter **605** are constructed similarly to combination rotary cutter and scorer unit **105** and together accomplish a corresponding objective. Essentially, rotary dough cutter and scoring unit **105** has been split into two units, with rotary scorer **600** being shown located upstream of rotary cutter **605** such that deformation scoring can take place before blunt-edge cutting. However, rotary scorer **600** could advantageously be located downstream of rotary cutter **605** so the upper skin is pre-stretched from the blunt-edge cutting prior to be scored. Certainly, if the dough was sliced along the sides, this would reduce the expansion at the score because, to at least a significant extent, expansion would occur along the side cuts. In any case, the rotary scorer **600** can be located longitudinally offset from rotary cutter **605** relative to or along the conveyor in accordance with the invention. Rotary scorer **600** includes a plurality of circular wheels **610-613**, which are configured to score dough sheet **300** in a manner directly corresponding to wheels **130-133**, and rotary cutter **605** includes a plurality of circular blades **615-619**, which, again in this example, are preferably blunt-edge blades or cutters and are configured to cut dough sheet **300** into dough strips **305** just like blades **125-129**. In the embodiment, wheels **610-613** again are smaller in size (diameter and thickness) than blades **615-619**.

(24) Each of rotary scorer **600** and rotary cutter **605** includes a driveshaft **620**, **621** to which wheels **610-613** or blades **615-619** are fixedly coupled, respectively. Accordingly, rotation of driveshafts **620** and **621** results in rotation of wheels **610-613** and blades **615-619**. Driveshafts **620** and **621** are aligned perpendicular to direction **115**. Therefore, the axes of rotation for rotary scorer **600** and rotary cutter **605** (labeled **625** and **626**, respectively) are also aligned perpendicular to direction **115**. As described above, rotary cutter **605** can be configured such that contact with conveyor belt **110** or dough sheet **300** causes rotary cutter **605** to rotate in a direction **630** as conveyor belt **110** travels in direction **115**. Alternatively, rotary cutter **605** can be driven using one or more motors. In the case of rotary scorer **600** of this embodiment, shaft **620** is positively driven to rotate as detailed further below.

(25) Each of rotary scorer **600** and rotary cutter **605** is supported above conveyor belt **110** by a support structure **635**. The details of support structures **635** are not important to the present invention. However, it can be seen that each support structure **635** includes identical left and right supports **640**. Each support **640** has a base **645** and a pair of columns **650**, which support a housing **655** receiving the ends of driveshafts **620** and **621**. If one or more motors are employed, the same would be located in a respective housing **655**, which in the case for at least the driveshaft **621** of rotary scorer **600**.

(26) Like rotary cutter **105**, rotary scorer **600** and rotary cutter **605** are used in the production of scored dough pieces, such as dough piece **400**, where the scores create weak zones which remain

through freezing, thawing and proofing, but which cause the bread dough to open up along the score line upon heating. Both embodiments achieve this goal by combining the cutting of the bread dough with deformation and pre-stressing. Again, the scoring deforms the dough in such a way that the score remains intact during later production steps and all the way to receipt of the bread dough by end users. That is, the score remains intact through any partial proofing and freezing by the producer, as well as both thawing and proofing by the end user. However, upon being heated, such as by an end user, the bread dough opens up along the score line to create opening **504**.

(27) By offsetting driveshafts **620** and **621**, it becomes possible to provide for vertically shifting of at least rotary scorer **600**. By way of example, an actuator **700** can be positioned between base **645** and each housing **655** for this purpose. In the case of a linear actuator, a cylinder **710** can be mounted on base **645** and a piston rod **720** can extend to housing **655**, with columns **650** guiding housing **655** for selective vertical movement relative to conveyor belt **110**. This arrangement advantageously enables more precise control of the depth of each of the scores into the body of the dough, thereby accommodating varying dough thicknesses and formulations. Furthermore, the raising and lowering of rotary scorer **600** (i.e., wheels **610-613**) provides for the ability to create discontinuous score lines in the dough strips. Of course, the invention is not limited to the use of symmetrical or circular deformation tools forming just one or more elongated linear scores per dough piece, but rather other deformation scoring tools can also be employed.

(28) Based on the above, it should be recognized that, in accordance with the present invention, mass production with deformation scoring of an un-proofed dough prior to freezing advantageously results in later thawed, proofed and baked dough pieces with which are more uniform in functional and aesthetic attributes comparable to dough products which are scored after proofing such that the present invention provides pre-stressed, scored dough pieces where the score remains intact through freezing, thawing and proofing, as well as apparatuses and methods for producing the scored dough pieces. While certain preferred embodiments of the present invention have been set forth, it should be understood that various changes or modifications could be made without departing from the spirit of the present invention. In general, the invention is only intended to be limited by the scope of the following claims.

Claims

1. A method for producing frozen un-proofed dough pieces with scores, the method comprising: transporting an un-proofed dough as a dough sheet with a conveyor, the dough having a body, a lower skin in contact with the conveyor and an upper skin opposite the lower skin; cutting the dough with a plurality of blades configured to cut the dough into a plurality of dough strips; scoring the dough using a blunt deformation tool to produce a functional score for each of the plurality of dough strips by drawing the upper skin into the body to a depth above the lower skin while deforming and pre-stressing the body without cutting the dough utilizing the blunt deformation tool; cutting the plurality of dough strips to form a plurality of un-proofed dough pieces; and freezing the un-proofed dough pieces.
2. The method of claim 1, wherein the score is formed either before or after cutting the plurality of dough strips.
3. The method of claim 1, wherein a plurality of scores is formed simultaneous with cutting the dough sheet to form the plurality of dough strips.
4. The method of claim 1, wherein the plurality of blades employed for cutting the dough sheet constitutes a plurality of blunt-edge cutters, each having a dough engaging portion configured to press the upper skin toward the lower skin to pinch the upper and lower skins together while cutting to form first and second dough strip sides for each of the plurality of dough strips.
5. The method of claim 1, wherein each deformation tool configured to produce a score is a wheel fixed to a first driveshaft, each of the plurality of blades configured to cut the dough sheet is a

circular cutter fixed to a second driveshaft, and the first driveshaft is located offset from the second driveshaft along the conveyor.

6. The method of claim 5, wherein the second driveshaft is motor driven to rotate and is driven at a position upstream of the first driveshaft.

7. The method of claim 6, further comprising vertically adjusting the first driveshaft relative to the conveyor.

8. The method of claim 1, wherein the score is configured to form an opening during baking of the dough piece.

9. The method of claim 1, wherein the score is parallel to a longitudinal axis of a respective one of the plurality of dough strips.

10. The method of claim 7, further comprising allowing the dough to rebound from the scoring prior to freezing the dough.
